UNIT 1 · STANDARD 5.MD.C.5

Math is Explaining and Sharing

Lesson 1-4

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.
Language Objective: I can explain my thinking with clear math vocabulary using the words argument, conjecture, and cubic units.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

argument | conjecture | counterexample | volume | cubic unit

👆 *Tap any word to see what it means and a picture.*

1. ____________ — A chain of reasons — equations, drawings, or words — used to defend your thinking.

2. ____________ — A mathematical statement you believe is true but have not yet proven.

3. ____________ — A single example that shows a statement is not always true.

4. ____________ — The amount of space a solid figure takes up, measured in cubic units.

5. ____________ — A cube with edges 1 unit long — the unit used to measure volume.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

How can the group determine the amount of soil needed, and how can they check that their answer is accurate?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **argument** — A chain of reasons — equations, drawings, or words — used to defend your thinking.
  argumento — Una cadena de razones — ecuaciones, dibujos o palabras — que usas para defender tu pensamiento.
- [ ] **conjecture** — A mathematical statement you believe is true but have not yet proven.
  conjetura — Una afirmación matemática que crees verdadera pero que aún no has probado.
- [ ] **counterexample** — A single example that shows a statement is not always true.
  contraejemplo — Un solo ejemplo que muestra que una afirmación no siempre es verdadera.
- [ ] **volume** — The amount of space a solid figure takes up, measured in cubic units.
  volumen — La cantidad de espacio que ocupa una figura sólida, medida en unidades cúbicas.
- [ ] **cubic unit** — A cube with edges 1 unit long — the unit used to measure volume.
  unidad cúbica — Un cubo cuyas aristas miden 1 unidad — la unidad que se usa para medir el volumen.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

To find how much soil is needed, find the ____ of the garden beds. Because volume covers three dimensions, label the answer in ____ feet. Finding the volume two different ways and getting matching totals shows the solution is ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Three large boxes hold 240 × 3 = 720 cubic inches, and two jumbo boxes hold 400 × 2 = 800 cubic inches, so 800 > 720.

Evidence: A strong answer calculates both totals (720 and 800) and compares them to recommend the jumbo boxes. A weak answer picks an option without computing and comparing both volumes.

Reasoning: This evidence connects the definition of argument to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **To find how much soil is needed, find the ____ of the garden beds. Because volume covers three dimensions, label the answer in ____ feet. Finding the volume two different ways and getting matching totals shows the solution is ____.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.